

Faculty of Engineering & Technology

Computer Science Department

Software Engineering- COMP433

First Semester 2025 – 2026

Home Plants Shop - Group 8

Final Report

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Section: 4.

Date: 28/01/2026.

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Chapter 1: Project Planning and Management

1.1 Names of Editors/writers of the Report

All team members participated equally in writing and editing this report.

1.2 Business Title

Home Plants Shop : to select, buy and home deliver different types of Home Plants

1.3 Group Name: Group Number

G8

1.4 Name of Students (Members)

Member 1: Jana Quttosa	Number: 1210331
Member 2: Khadija Absi	Number: 1220458
Member 3: Duaa Naji	Number: 1220336
Member 4: Heba Abu Taha	Number: 1221211
Member: Enas Qutit	Number: 1210236

1.5 Role of each member

Member 1: : Jana Quttosa	Role: Project Manager
Member 2: Heba Abu Taha	Role: Technical Architect
Member 3: Enas Qutit	Role: Quality Assurance (QA)
Member 4: Duaa Naji	Role: Requirement Engineer
Member 5: Khadija Absi	Role : programmer

1.6 Project Management Strategy

To ensure the project's success and maintain effective communication, our team held weekly online meetings through Google Meet to track progress, raise different issues, plan the next work, and evaluate the results of work done so far.

Throughout the work at all phases, major decisions were taken by voting by all team members. This emphasized shared responsibility in directing the project and made sure that decisions represented the team's overall judgment.

For our Home Plants shop software development, agile model was used. It is flexible and supports frequent changes in requirements considering the customer feedback. Additionally, it supports Iterative delivery that gives a great opportunity to release the minimal viable feature early, collect feedback, and improve the system out of that.

1.7 Project Manager Report

Team Contributions:

[**Jana Quttosa: Role: Project Manager]:**As the project manager, I oversaw the entire lifecycle of the project, ensuring smooth coordination among team members and maintaining clear, consistent communication. I took charge of tracking progress and meeting deadlines, while also stepping in to support teammates who encountered setbacks. Additionally, I played a key role in organizing project documentation and actively participated in crafting and presenting the final project deliverables.

[Duaa Naji, Role: Requirement Engineer (Product Owner)]:As the Requirement Engineer (Product Owner), Duaa was responsible for leading the requirements elicitation process with the support of other team members. she focused on identifying, analyzing, and documenting user needs to ensure that requirements were clear, complete, and aligned with the project objectives. she worked closely with the team to clarify requirements, resolve ambiguities, and maintain consistency between user requirements and system design artifacts throughout the project.

[Heba Abu Taha , Role: Technical Architect]: As the technical architect, Heba was active throughout all project phases. She participated in the discussions of all design diagrams and provided constructive feedback to ensure correctness and consistency across them. She helped in clearing up design issues and suggested useful solutions. She also contributed in preparing the project's presentation and final report.

[Khadija Absi , Role: Programmer]: As the programmerKhadija was responsible for contributing to the design of essential system features that support user interaction. She collaborated with the team to translate requirements into functional components and took part in reviewing and refining the system diagrams to ensure usability and consistency. She also contributed in preparing the project's presentation and final report.

[Enas Qutit, Role: Quality Assurance]: As the QA lead, Enas ensured the clarity, consistency, and correctness of all project deliverables. She reviewed documents and diagrams to meet system goals and quality standards. Sara also supported the team by providing feedback 3 | Page and verifying final outputs. She also contributed in preparing the project's presentation and final report.

Challenges Faced:

Although the UML diagrams and software engineering concepts were introduced in the lectures, the biggest challenge was that the provided slides examples were often small and limited. In order to ensure full understanding, each member searched across different resources outside the course materials like course book, visual-paradigm website and more. So, applying these concepts to our project required a much deeper understanding, high attention to details, and careful planning to do tasks as we learned. Also, we faced some difficulty in maintaining consistency and correctness across all diagrams that needed to understand the full interactions between components and actors. Last challenge was the time; it was a recurring challenge as we had to balance this project with other assignments and deadlines from different courses.

Project Success:

In my opinion, our project was successfully completed mainly because of the strong teamwork and continuous collaboration. Every part of the project was fully discussed by all team members, where each one shared their opinion and evaluated the work until we reached a decision we all agreed on, based on proper research and understanding. We also carefully considered Doctor Adel ’s feedback after each phase, which helped us improve by addressing important notes from the lectures. The team’s commitment, openness to discussion, and our ability to learn together were the real reasons for project success.

1.8 Group members report

Members Name: Jana Quttosa

I consider our project a well-executed and collaborative success that met all requirements and delivered a robust system design. My contributions spanned multiple areas: I authored both user and system requirements, reviewed and finalized them to ensure precision and completeness, and led the creation of the effort and time estimation table. I also developed the scenario analysis for the "placing an order" use case and reviewed the actor analysis and their descriptions. In addition, I helped finalize the overall system use case diagram and wrote a detailed description of my assigned use case, aligned with its scenario. I actively participated in discussions around the system activity modeling and personally created the activity diagram for my use case. My involvement extended to reviewing the system's class modeling and design goals, finalizing the component diagram, and contributing to architecture diagram discussions. I also created the sequence diagram for my use case and led the development of the deployment diagram. Overall, my role combined leadership, technical writing, diagram creation, and collaborative review to support the project's success.

Member's Name: Duaa Naji

I evaluate the project as a successful outcome of effective requirements engineering and team collaboration. As the Requirement Engineer (Product Owner), my main responsibility was to lead the requirements elicitation activities and ensure that user needs were accurately captured and documented. I worked with team members to analyze requirements, refine them for clarity and completeness, and maintain consistency between user requirements and system requirements. I also participated in discussions across different project phases to ensure alignment between requirements and the produced design models. My contributions helped establish a clear requirements foundation that supported the analysis and design of the system.

Member's Name: Heba Abu Taha

As a technical architect, I think the project is successful. It delivered a structured and functional system supported by clear architectural planning and consistent design models. The main challenges I faced involved maintaining consistency across different design models to align the overall design and ensure smooth system integration. I contributed across all phases by leading key tasks and actively participating in others. I led the user requirements documentation. I participated in the system requirements discussion to ensure correctness and consistency. I reviewed the effort and time estimation calculations. Additionally, as individual tasks, I wrote a detailed scenario for the use case. I also wrote the detailed description for it. I drew the activity diagram model for the same use case. I led the actor analysis process to define the system actors. I participated in use case modeling discussion and ensured it captured system functionalities as seen by users. I finalized the overall activity model. I led the system class diagrams by creating both the analysis and the detailed class diagrams. Additionally, I drew a sequence diagram for the use case. I contributed to defining system design goals. I reviewed the system components and finalized the system architectural model. Finally, I reviewed the deployment diagram.

Member's Name: Khadija Absi

I evaluate the project as a successful result of effective teamwork and clear coordination between analysis, design, and implementation phases. As the Programmer, my main responsibility was to implement the system components according to the specified requirements and design models. I translated use cases and sequence diagrams into functional system logic, ensured proper integration between system modules, and resolved implementation-related issues. I also collaborated with team members to verify that the implemented features met the defined requirements and supported the overall system objectives. My contributions helped ensure that the system was functional, reliable, and aligned with the project design and requirements.

Member's Name: Enas Qutit

I think our project was successful and well organized. As the QA leader, my main job was to check the quality of our work and make sure everything is clear and correct.My contributions included checking the User Requirements and System Requirements to ensure they are complete, clear, and match the project scope. I reviewed the scenarios and use case descriptions to make sure they follow the same steps and assumptions.I also reviewed the actor analysis and actor descriptions to ensure each actor has a clear role, with no missing or repeated roles. I helped review the effort and time estimation to confirm the numbers and assumptions are reasonable.For the design models, I reviewed the activity diagrams, sequence diagrams, and class diagrams to ensure they are correct and match the written requirements. I also helped improve the report by fixing wording, keeping the same terms, and checking the format.

Overall, my role was focused on quality checking, consistency, and giving feedback to improve the final report.

Chapter 2: Requirement Elicitation, Analysis and Modelling

2.1 Requirement statement/Business description

Modern individuals and families often struggle to find time to visit physical plant nurseries due to busy schedules, limited store hours, or lack of expert guidance on plant care. Many customers are unsure which plants suit their homes, lighting conditions, or lifestyles Home Plants Shop addresses this challenge by providing a centralized, reliable online platform where users can easily explore a wide variety of home plants, access care information, compare options, and receive expert support—reducing the effort, time, and uncertainty associated with traditional plant shopping.

Home Plants Shop is an online home plants shop designed to meet the needs of plant enthusiasts, homeowners, and beginners alike. Operating through a responsive website, the platform allows users to conveniently browse, select, and purchase indoor plants, outdoor plants, pots, soil, fertilizers, and plant care accessories. The system offers a smooth shopping experience, multiple payment options, and detailed product descriptions including care instructions, sunlight requirements, watering schedules, and customer reviews, enabling users to make informed decisions from the comfort of their homes.

The primary users of the system include customers, inventory workers, support agents, and managers. Customers can search for plants, filter products based on criteria such as plant type or care level, manage wish lists, place orders, and receive assistance through a chatbot or live support agents. Inventory workers are responsible for managing the product catalog by adding new plants, updating stock levels, and maintaining accurate plant information. Support agents handle customer inquiries related to plant care, orders, and deliveries, especially those escalated from the chatbot. Managers oversee overall shop operations, monitor performance, and generate detailed sales and inventory reports.

This system addresses the growing demand for convenient, educational, and trustworthy online plant shopping experiences. It aims to improve accessibility for busy individuals and beginner plant owners while streamlining inventory management and customer support processes. The

platform is designed following core software engineering principles such as low coupling, high cohesion, and usability, ensuring that each system component operates efficiently, remains easy to maintain, and provides a clear and user-friendly experience.

2.2 USER and SYSTEM requirements

1. **USER REQUIREMENTS:**

Lead: Jana

Contributors: Enas: **co-leading,** Duaa**: reviewing,** Heba**: discussing,** Khadija**: finalizing**

- Customers shall be able to browse home plants.
- Customers shall be able to view plant information.
- Customers shall be able to add plants to a shopping cart.
- Customers shall be able to provide their personal information.
- Customers shall be able to complete payment.
- Customers shall be able to track their orders.
- Customers shall be able to receive after-service support.

2. **SYSTEM REQUIREMENTS**

Lead: Enas

Contributors: Khadija: **co-leading,** Heba**: reviewing,** Duaa**: discussing, Jana: finalizing**

1. **System requirements for browsing and searching plants:**

1.1. The system shall categorize plants into types such as small indoor, medium-sized decorative, and low-maintenance.

1.2. The system shall enable searching and filtering by plant name, category, or price range.

1.3. The system shall update plant availability in real-time to show only in-stock items.

2. **System requirements for plant details:**

2.1. The system shall provide plant name, price, picture (in different seasons for each plant), and care instructions including watering and sunlight.

2.2. The system shall update plant details to reflect current stock and price changes.

3. **System requirements for selecting plants and shopping cart:**

3.1. The system shall allow addition of plants to a cart and adjustment of quantities.

3.2. The system shall calculate total cost based on selected plants and quantities.

3.3. The system shall remove items from the cart upon request.

3.4. The system shall notify if a selected plant becomes unavailable before checkout.

4. **System requirements for providing personal information and selecting delivery method**

4.1 The system shall collect customer name and phone number during checkout.

4.2 The system shall allow customers to choose between home delivery and store pickup.

4.3 If home delivery is selected, the system shall require and validate the delivery address.

4.4 The system shall store customer information and the selected delivery method securely for order processing.

5. System requirements for payment:

5.1. The system shall offer payment options including card, digital wallet, or cash on delivery.

5.2. The system shall validate payment details based on the chosen method.

5.3. The system shall confirm successful payment before order completion.

5.4. The system shall allow retry with a different method if payment fails.

6. System requirements for order confirmation and delivery:

6.1. The system shall confirm orders and assign a unique order ID.

6.2. The system shall provide real-time order status updates, displaying each status with a distinct color, for example: Preparing is yellow, Out for Delivery is blue, and, Delivered is green.

6.3. The system shall notify customers of status changes.

6.4. The system shall reduce inventory levels after successful order confirmation.

6.5. The system shall calculate standard delivery fees.

6.6. The system shall apply discounts if total order value exceeds 150 NIS.

6.7. The system shall add extra charges for express same-day delivery.

6.8. The system shall include fees in the total cost before payment.

7. System requirements for after-service support:

7.1. The system shall provide plant care tips upon request.

7.2. The system shall enable customers to submit questions and receive responses including an FAQ section.

7.3. The system shall record and track all customer support interactions to enable follow-up and resolution.

7.4. The system shall provide a WhatsApp Link to answer the customer questions.

2.3 Scenario Analysis

Scenario 1: Add a New Plant to the Catalog [Enas Qutit]

- **Initial Assumption**

The inventory worker is logged into the system and has permission to add plants to the catalog.

- **Normal Flow**

The inventory worker chooses “Add New Plant” from the inventory menu. The system shows a form to add plant information. The inventory worker enters the plant name, chooses a category such as indoor or outdoor enters the care level price and available quantity then submits the form. The system checks that all required fields are filled correctly.after that the system checks if a plant with the same name and category already exists if no duplicate is found the system saves the plant information. Finally, the system shows a message that the plant was added successfully.

- **Alternative Flow**

Alternative 1: Category Selection via Search

Instead of selecting the category from the list the inventory worker searches for the category by typing its name the system shows matching categories the inventory worker selects one and continues normally this process is completed successfully. [Successful outcome]

- **Error Flow**

If the system finds an existing plant with the same name and category the system does not save the plant an error message is shown and the inventory worker can cancel the process or start again if a system or database error happens during submission the plant is not saved and the system asks the inventory worker to try again later in both cases the process is not completed. [Unsuccessful outcome]

- **System State on Completion**

If the process is successful, the new plant is saved in the inventory database and appears in the catalog. If the process fails, no data is saved and the system remains unchanged.

Scenario 2: Customer Places an Order [Heba Abu Taha]

- **Initial Assumption**

The customer is already registered and logged into the system. The platform is online and fully operational, ready for transactions. The customer has access to the plant catalog and can view available products.

- **Normal Flow**

The customer browses the catalog, selects an in-stock plant, and adds it to the shopping cart. Then, they proceed to checkout, enter shipping details, and select "Online Payment" as the payment method.

The payment system processes the transaction, authorizes the payment, and confirms the successful transaction. Once confirmed, the system updates the stock level of the selected plant and reserves it for the customer. An order confirmation message is displayed with details such as the order number, purchased items, and estimated delivery date.

- **Alternative Flow**

Alternative 1: Cash on Delivery

If the customer prefers not to pay online, they choose "Cash on Delivery" during the checkout process. The system bypasses payment authorization and immediately confirms the order.

The stock is reserved for delivery, and a confirmation message is displayed indicating that the order will be delivered once the customer pays in cash upon arrival.

- **Error Flow:**

If the payment system fails to authorize the transaction, the system cancels the order. The customer is notified with an error message stating that the payment was unsuccessful and the transaction could not be processed. The order is not recorded, and no changes are made to the stock levels.

If the payment method is invalid or the customer’s card is declined, the system asks the customer to retry with a different payment method. The system ensures that no items are shipped until payment is successfully completed.

- **System State on Completion:**

Upon successful order placement, the system records the order details, updates the stock, and sends a confirmation email to the customer.

If the payment fails or Cash on Delivery is selected, no order is recorded, and the stock remains unchanged. The customer is informed to retry with a different payment method or to select an alternative.

Scenario 3: Customer Tracks an Order [Individual Task – Duaa Naji]

- **Initial Assumption**

The customer is logged into the system and has previously placed at least one order.

- **Normal Flow**

The customer navigates to "My Orders" from the main menu. The system retrieves and displays a list of the customer's orders. The customer selects a specific order from the list. The system displays the order details, including the current order status (e.g., Preparing, Out for Delivery, Delivered). Successful outcome: Yes

- **Alternative Flow**

The customer clicks the order tracking link received via email. The system identifies the order using the embedded order ID and displays the order status directly, without requiring navigation through the account menu.

Successful outcome: Yes

- **Alternative Flow**

The customer has multiple orders placed on similar dates. The system displays additional details (order number, total price, order date) to help the customer identify the correct order.

Successful outcome: Yes

- **What Can Go Wrong?**

Error: The order ID cannot be retrieved or does not exist in the system. The system displays an error message indicating that the order is unavailable or invalid.

Successful outcome: No

Error: The customer attempts to access an order that does not belong to their account. The system denies access and displays an authorization error message.

Successful outcome: No

- **Other Activities**

Order information may be viewed by customer support staff, but not modified during tracking. The system may refresh order status automatically if the delivery state changes.

- **System State on Completion**

No system data is modified. Order tracking is a read-only operation. The system logs the access event for monitoring and auditing purposes.

Scenario 4: Customer Requests Customer Support [Khadija Absi]

- **Initial Assumption**

The customer is logged into the Plant Shop system and has access to customer support services. The platform is online and functioning normally.

- **Normal Flow – Chatbot Support**

The customer clicks the “Customer Support” button in the system. A chatbot window opens and prompts the customer to select an issue category, such as Delivery Issue, Damaged Plant, Payment Problem, or General Inquiry. The customer selects a category and provides a brief description of the issue, for example, “My plant arrived damaged.” The chatbot responds with automated guidance or troubleshooting steps based on the issue type.

If the issue is not resolved after a few exchanges, the system escalates the conversation to a human Support Agent. The Support Agent reviews the customer’s account and order history, then communicates directly with the customer to resolve the issue. Once the problem is resolved, the customer confirms and closes the chat. The system records the interaction in the database, linking it to the customer’s account.

- **Alternative Flow – Support via WhatsApp Link**

Instead of using the chatbot, the customer can choose the “Contact via WhatsApp” option. The system opens a WhatsApp chat window or redirects to the WhatsApp app, pre-filling a message with the customer’s account ID and issue type. The customer communicates directly with a Support Agent through WhatsApp, who responds and resolves the issue in real-time. Once resolved, the customer confirms, and the system records the interaction, noting that WhatsApp was used as the support channel.

- **Error Flow – Support Services Unavailable**

If the chatbot fails to load or the WhatsApp link is temporarily unavailable, the customer sees a message stating, “Customer Support is currently unavailable. Please try again later.” The customer cannot proceed with support at that time.

- **Other Activities**

While interacting with customer support, the customer can continue browsing plants or checking their account. All interactions, whether through the chatbot or WhatsApp, are logged and preserved for future reference.

- **System State on Completion**

The customer remains logged in. All support requests and related interactions are stored in the system database and linked to the customer's account, including timestamps, support channel used, issue type, and resolution status. If a human agent assisted, their ID is also recorded.

Scenario 5: Shop Manager Generates Sales Report [Individual Task - Jana Quttosa]

- **Initial Assumption**

The Shop Manager is authenticated and has admin level access.

- **Normal Flow**

The shop manager navigates to the Reports section and selects the Sales Report option. After that, the manager enters the required date range and plant category. The system then retrieves the matching sales data and displays a preview of the report. Finally, the shop manager downloads the report in PDF format.

- **Alternative Flow**

Alternative 1: Default Report

The shop manager navigates to the Reports section and selects the Sales Report option. Instead of choosing any filters, the manager skips the filter selection. The system then automatically generates a default report for the last 7 days and displays a preview of the report. Finally, the shop manager downloads the report in PDF format. **[Successful outcome]**

Alternative 2: Refine Filters After Preview

The shop manager navigates to the Reports section and selects the Sales Report option. After entering the date range and plant category, the system displays a preview of the report. The shop manager reviews the preview and decides to refine the filters to get more specific results. The system updates the report preview based on the new criteria, and the shop manager downloads the final report as a PDF. **[Successful outcome]**

- **Error Flow**

The shop manager navigates to the Reports section and selects the Sales Report option. The manager skips the filter selection, so the system automatically generates a default report for the last seven days. If no sales data matches the criteria, the system displays a “No records found” message. **[Unsuccessful outcome]**

- **System State on Completion**

Reports are generated and downloaded, system data remains unchanged.

2.4 Effort/Time estimation calculation: Modified

Lead: Duaa

Contributors: Jana: **co-leading,** Heba**: reviewing,** Khadija**: discussing, Enas: finalizing**

User Requirements	**Estimated effort**	**Estimated No. of developer**	**Total Effort (for one developer)**
UR1	2 pw	2	4 pw
UR2	1 pw	1	1pw
UR3	1 pw	2	2 pw
UR4	2 pw	2	4 pw
UR5	2 pw	1	2 pw
UR6	2 pw	1	2 pw
UR7	1 pw	2	2 pw
Total effort / avg	11 pw	11 /5 = 2.2 dev on avg needed	17 pw
Schedule time 30%	11 * 1.30 = 15 w (min time to complete)		17 * 1.30 = 23w (max time to complete)
Cost		Avg salary = 200 $	200 *23 =4600**$**
Profit margin **(min=10%,max=30%)**		**Min Cost**	4600* 1.10 = 5060 **$**
		Max Cost	4600* 1.30 = 5980 **$**

2.5 ACTORS analysis and their description

Lead: Khadija

Contributors: Jana: **co-leading,** Heba**: reviewing,** Duaa**: discussing, Enas: finalizing**

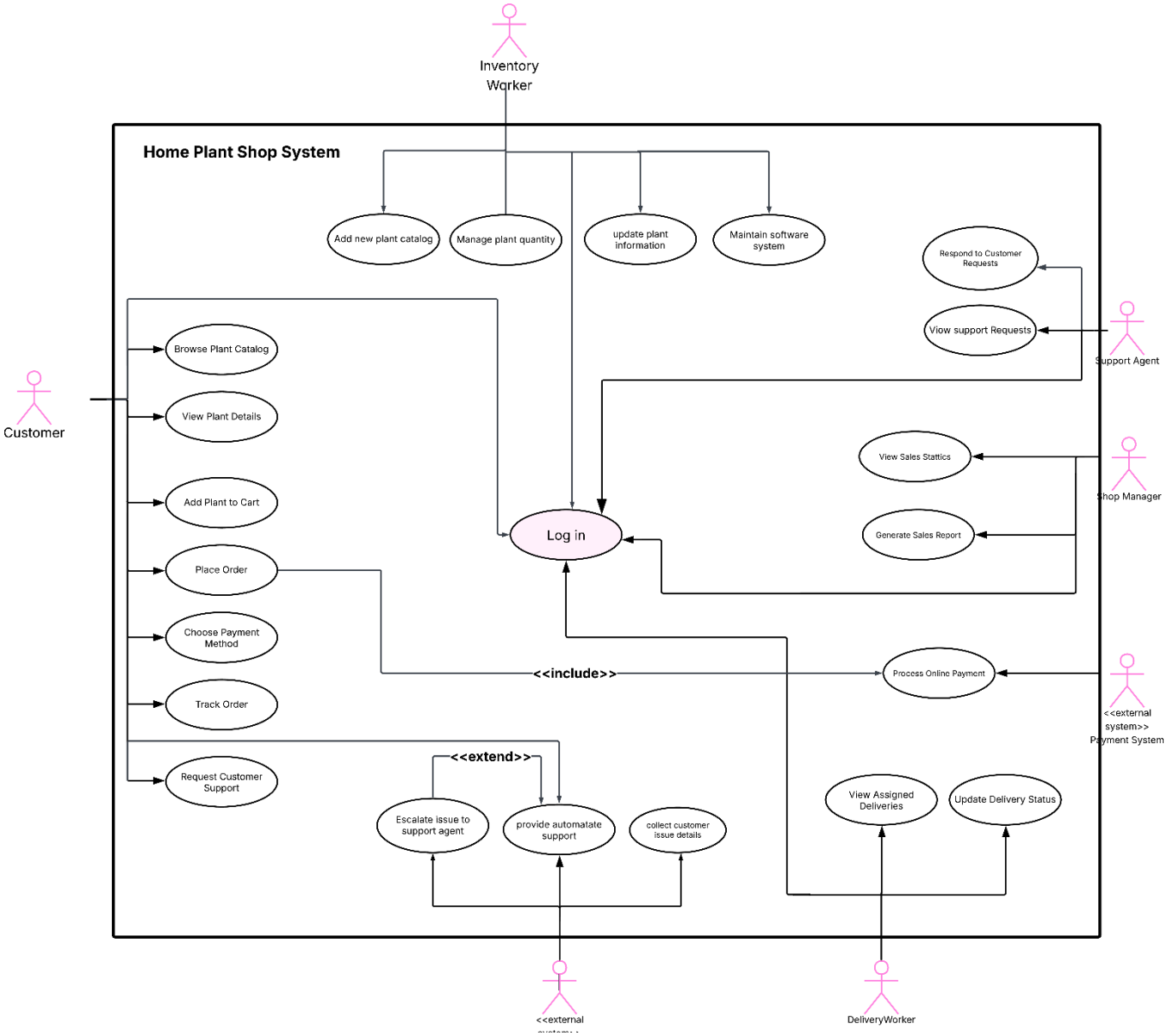
Actors	Description
Customer	The Customer is a registered user of the Home Plants Shop system. The customer can browse the plant catalog, view plant details, add plants to the shopping cart, place orders, choose a payment method, track existing orders, and request customer support. The customers can only access and manage their own orders and account information.
Inventory Worker	The Inventory Worker is a staff member responsible for managing the plant inventory. This actor can add new plants to the catalog, update plant information such as price and quantity, and ensure that plant data is accurate and up to date. The inventory worker does not handle customer orders or payments.
Shop Manager	The Shop Manager is an administrative user with higher privileges. This actor can generate sales reports, monitor system performance, and view overall sales statistics. The shop manager does not directly add plants or place orders but supervises business operations.
Support Agent	The Support Agent is responsible for handling customer support requests that cannot be resolved automatically. The support agent can view customer issues, order details, and provide assistance or resolutions through support channels.
Payment System	The Payment System is an external system that processes online payments. It validates payment details, authorizes transactions, and returns success or failure responses to the Home Plants Shop system.
Delivery Worker	The Delivery Worker is responsible for delivering plant orders to customers. This actor interacts with the Home Plants Shop system to view assigned deliveries and update the delivery status (such as *Out for Delivery* or *Delivered*). The Delivery Worker does not modify order

	details or payment information and only updates delivery-related information.
Chatbot	The Chatbot is an automated component of the Home Plants Shop system that provides initial customer support. It interacts with customers by answering frequently asked questions, guiding them through common issues, and collecting basic problem information. If the chatbot cannot resolve the issue, it escalates the request to a human support agent. The chatbot does not make decisions independently and operates based on predefined rules and responses.

2.6 USE-CASE diagram.

Lead: Jana

Contributors: Khadija: **co-leading,** Enas**: reviewing,** Heba**: discussing,** Duaa**: finalizing**



Link : USE-CASE diagram

2.7 USE-CASE Specifications

1. Add a New Plant to the Catalog [Enas Qutit] : Modified

2. Section	Description
Scenario Title	Add a new plant to catalog
Actor	Inventory Worker
Initial Assumption (Precondition)	The inventory worker is logged into the system.
Normal Flow	1. The inventory worker selects the option to add a new plant. 2. The worker enters the plant name as free text. 3. The worker selects the plant category from a predefined list (Outdoor Plant, Indoor Plant, etc.). 4. The worker enters the brand name. 5. The worker enters the age suitability for the plant. 6. The worker enters the plant price. 7. The system validates that all required fields are filled and values are valid. 8. The system checks for duplicate plant using name and brand. 9. If no duplicate is found, the system adds the plant to the catalog. 10. The system displays a summary and a success message.
Alternative Flow (Successful Output)	If the required category is not found, the worker selects “Other” and enters the category details manually. The system accepts the input and continues successfully.
Error Flow 1 (Unsuccessful Output)	If the category list fails to load due to a database error, the system displays: “Unable to load categories. Please try again later.” The form cannot be submitted and the plant is not added.
Error Flow 2 (Unsuccessful Output)	If the system detects that the plant already exists in the catalog, it displays: “The plant is a duplicate.” The plant is not added.

Other Activities	The plant catalog is always available, but the new plant is not displayed until the addition process is completed successfully.
System State on Completion (Postcondition)	The worker remains logged in. The new plant is saved in the inventory database, displayed in the catalog, and a log entry is created with the time, plant details, and worker ID.

3. Customer Places an Order [Heba Abu Taha]

Actor(s):	Customer, Payment System
Description:	A customer places an order for plants via the online shop. The system only allows adding items to the cart if they are available in stock. Once the customer proceeds to checkout, the payment system processes the transaction. Upon successful payment, the order is confirmed, and inventory levels are automatically updated.
Pre-conditions:	1.Customer is registered and logged into the system. 2.The desired plants are available in the inventory with sufficient stock quantity to fulfill the customer's order.
Sequence/Flow of Events:	1.Customer browses the plant catalog. 2.Customer selects in-stock plants and adds them to the shopping cart. 3.Customer reviews the cart and proceeds to checkout. 4.Customer enters payment details. 5.Payment System processes the transaction. 6.If payment succeeds, the system confirms the order and updates the stock. 7.If a plant is unavailable or payment fails, the system notifies the customer (alternative path).
Data:	1.Plant details (ID, name, quantity, price) 2.Customer information 3.Order and payment information 4.Inventory levels
Stimulus/Trigger:	User command issued by the system.
Post-conditions / Response:	1.Order is recorded, and a confirmation is shown to the customer. 2.Inventory is automatically updated to reflect the purchase.

	3.If payment fails, the order is not recorded, and no changes are made to the stock. An error message is displayed to the customer indicating the payment failure.
Comments:	1.System restricts out-of-stock plants from being added to the cart. 2.Payment System ensures secure handling of payment data. 3.Customer Support Staff handle questions about stock or payment issues.

4. Customer Tracks an Order [Duaa Naji]

Use case: Customer Tracks an Order	
Actors	Primary Actor: Customer Secondary Actor: Customer Support Staff (view-only)
Description	This use case allows a customer to track the status and details of a previously placed order. The customer can access order tracking either through their account or via a tracking link received by email. The system displays the current order status without allowing any modification to the order data.
Pre-conditions	• The customer has previously placed at least one order. • (For account-based tracking) The customer is authenticated and logged into the system. • OR (For email-based tracking) The customer has received a valid tracking link via email.
Trigger	The customer requests to track an order.
Sequences/ flow of events (Normal flow)	1. The customer navigates to "My Orders" from the main menu. 2. The system retrieves and displays a list of the customer's orders. 3. The customer selects a specific order from the list. 4. The system retrieves the order information. 5. The system displays the order details, including: • Order number and order date • Items ordered with quantities • Delivery address • Total order price • Current order status (e.g., Preparing, Out for Delivery, Delivered) • Order status history timeline with timestamps • Estimated delivery date (if available)

Alternative flow	**A1: Track Order via Email Link** 1. The customer clicks the order tracking link received via email. 2. The system validates the tracking token embedded in the link. 3. The system identifies the order using the embedded order ID. 4. The system displays the order status directly without requiring the customer to log into their account. **A2: Multiple Similar Orders** 1. When multiple orders exist with similar dates or details, the system displays additional order information including order number, order date, and total price. 2. The customer uses this information to identify and select the correct order. 3. The system displays the selected order details as described in the Main Flow.
Error flow	**E1: Invalid or Non-existent Order ID** • **Condition:** The system cannot retrieve the order using the provided order ID. • **System Response:** The system displays an error message: "Order not found. The order ID is invalid or the order does not exist in the system." • **Outcome:** Unsuccessful **E2: Unauthorized Order Access** • **Condition:** The customer attempts to access an order that does not belong to their account. • **System Response:** The system denies access and displays an authorization error message: "Access denied. You do not have permission to view this order." • **Outcome:** Unsuccessful **E3: Expired or Invalid Tracking Link**

	• **Condition:** The tracking link from the email has expired or contains an invalid/corrupted token. • **System Response:** The system displays an error message: "This tracking link has expired or is invalid. Please log into your account to track your order, or contact customer support for assistance." • **Outcome:** Unsuccessful **E4: System Temporarily Unavailable** • **Condition:** The system cannot retrieve order data due to technical issues, database errors, or maintenance. • **System Response:** The system displays a message: "Order tracking is temporarily unavailable due to system maintenance. Please try again in a few minutes." • **Outcome:** Unsuccessful
Post conditions/ Response	• The order tracking information is successfully displayed to the customer. • No order data is modified during this operation (read-only operation). • The system logs the order tracking request with timestamp and user information for monitoring and auditing purposes.
Comments	Business Rules / Notes: • Customers can only view orders associated with their own accounts when accessing via account login. • Order tracking does not allow editing or cancellation of orders. This is a view-only operation. • Customer support staff may view order tracking information for customer assistance purposes but cannot modify any order data. • Order tracking links sent via email expire after 90 days from the date of order placement. • The system automatically refreshes order status when the delivery state changes in real-time or near real-time.

	• Order status updates from delivery partners may take up to 15 minutes to reflect in the system. • Guest users accessing via email tracking link see the same order details as logged-in customers. • Order status definitions: ○ Preparing: Order is being processed and prepared for shipment ○ Out for Delivery: Order is with the delivery courier and in transit ○ Delivered: Order has been successfully delivered to the customer ○ Cancelled: Order has been cancelled (if applicable) • The system maintains a complete history of all status changes with timestamps for customer reference.

5. Customer Requests Customer Support [Khadija Absi]

Field	Description
Actors	Customer, Chatbot, Support Agent
Description	A customer requests help through the platform for issues such as order problems, delivery issues, damaged products, or payment questions. The system first uses a chatbot to provide automated guidance. If the issue is not resolved, a human support agent takes over.
Pre-conditions	1. Customer is logged into the system. 2. Customer clicks on the “Customer Support” option.
Sequence / Flow of Events	1. Customer opens the support menu. 2. System opens the chatbot window. 3. Chatbot asks customer to choose issue type. 4. Customer selects or types the issue (e.g., “plant arrived damaged”). 5. Chatbot provides automated troubleshooting. 6. If unresolved after several messages, issue is escalated. 7. Support agent assists and resolves the issue. 8. System logs the entire interaction.
Alternative Flow	- Customer chooses “Support Form” instead of chat. - Customer fills issue type and description. - Support agent replies later via email or message.
Data Used	- Customer account info - Order history - Issue type and description - Support conversation log
Stimulus / Trigger	Customer clicks the “Customer Support” button or submits a support form.
Post-conditions / Response	1. Customer receives support. 2. Support ticket is created and stored. 3. System logs all details including resolution status. 4. Agent ID stored if a human assisted.
Comments	Only logged-in users can access personalized support. All interactions are securely logged.

6. Shop Manager Generates Sales Report [Individual Task - Jana Quttosa]

Field	Description
Actor	Shop Manager
Description	The Shop Manager accesses the admin dashboard to generate a sales report by navigating to the Reports section. The manager may specify filter criteria such as date range and plant category, or allow the system to generate a default report. The system retrieves the relevant sales data, displays a report preview, and allows the Shop Manager to download the report in PDF format.
Pre-conditions	1. The Shop Manager is authenticated and logged into the system. 2. The Shop Manager has admin-level access rights. 3. Sales data exists in the system (otherwise, an informative message is shown).
Sequence / Flow of Events	1. The Shop Manager navigates to the "Reports" section of the admin dashboard. 2. The system displays available report options. 3. The Shop Manager selects "Sales Report." 4. The system prompts the Shop Manager to enter filter criteria (date range, plant category). 5. The Shop Manager enters the required filters and submits the request. 6. The system retrieves matching sales data based on the entered criteria. 7. The system displays a preview of the sales report in tabular format. 8. The Shop Manager clicks "Download." 9. The system generates and downloads the sales report in PDF format. 10. The system displays a confirmation message indicating successful report generation.
	Alternative Flow 1: Default Report (Successful Outcome) 1. The Shop Manager navigates to the "Reports" section and selects "Sales Report." 2. The Shop Manager skips selecting any filter criteria. 3. The system automatically generates a default sales report for the last 7 days. 4. The system displays a preview of the default report.

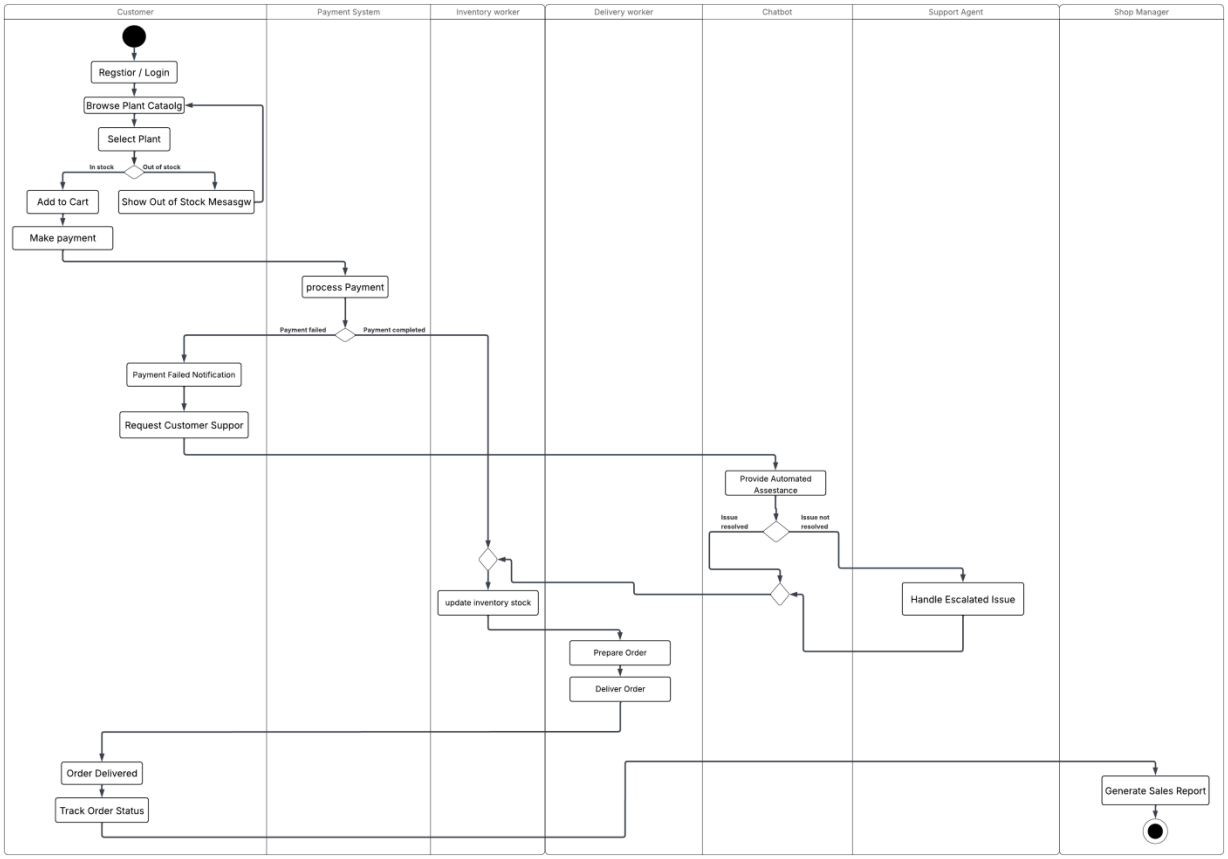
	5. The Shop Manager downloads the report in PDF format. 6. The system displays a confirmation message indicating successful download.
	Alternative Flow 2: Refine Filters After Preview (Successful Outcome) 1. The Shop Manager navigates to the “Reports” section and selects “Sales Report.” 2. The Shop Manager enters an initial date range and plant category. 3. The system displays a preview of the generated report. 4. The Shop Manager reviews the preview and decides to refine the filter criteria. 5. The Shop Manager updates the filters and resubmits the request. 6. The system updates and displays a new preview based on the refined criteria. 7. The Shop Manager downloads the final report in PDF format. 8. The system displays a confirmation message indicating successful report generation.
	Error Flow (Unsuccessful Outcome) 1. The Shop Manager navigates to the “Reports” section and selects “Sales Report.” 2. The Shop Manager skips filter selection, triggering the system to generate a default report for the last 7 days. 3. If no sales data matches the criteria, the system displays the message:“No records found for the selected criteria.” 4. No report is generated or downloaded.
Data	Sales records (plant names, quantities sold, prices, categories, timestamps).
Stimulus / Trigger	The Shop Manager selects the “Sales Report” option from the Reports section in the admin dashboard.
Post-conditions / Response	1. If sales data exists, the sales report is successfully generated, downloaded in PDF format, and a confirmation message is displayed. 2. If no sales data exists, no report is generated and an informative message is shown.

System State on Completion	Reports are generated and downloaded. System data remains unchanged.
Comments	The Shop Manager must have appropriate access permissions to generate and download sales reports.

2.8 ACTIVITY diagram: Modified

Lead: Heba

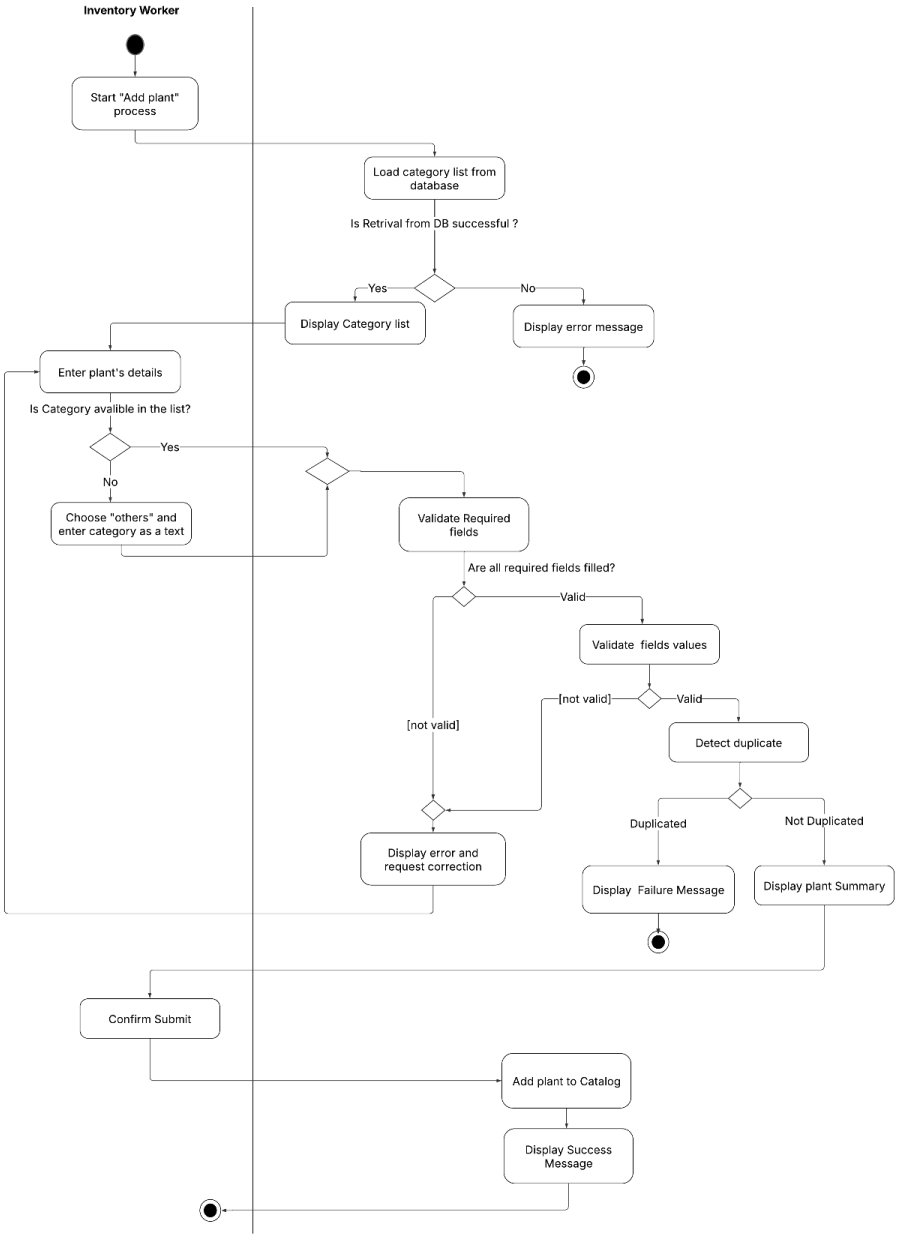
Contributors: Duaa: co-leading, Jana: reviewing, Enas: discussing, Khadija: finalizin



Link : [ACTIVITY diagram]

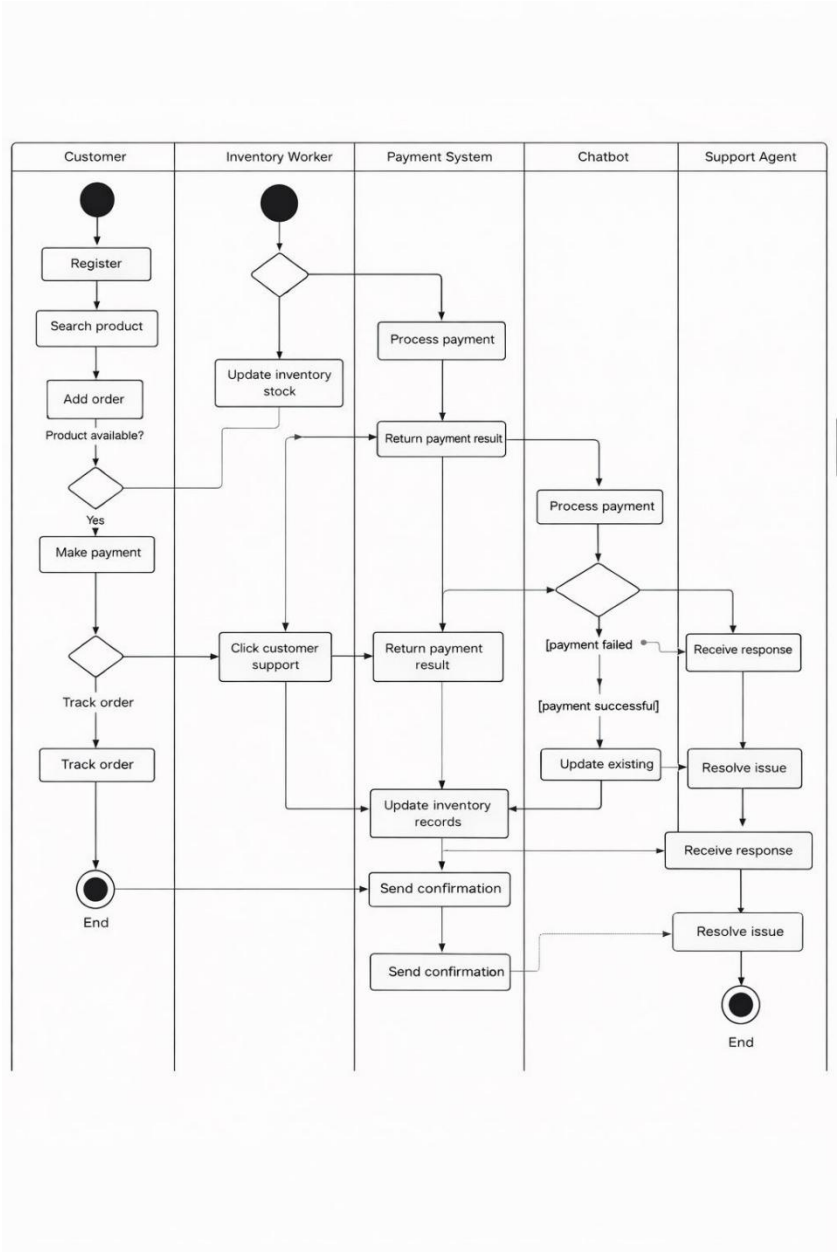
2.9 Instance Activity diagrams

1. Add a New Plant to the Catalog [Enas Qutit]



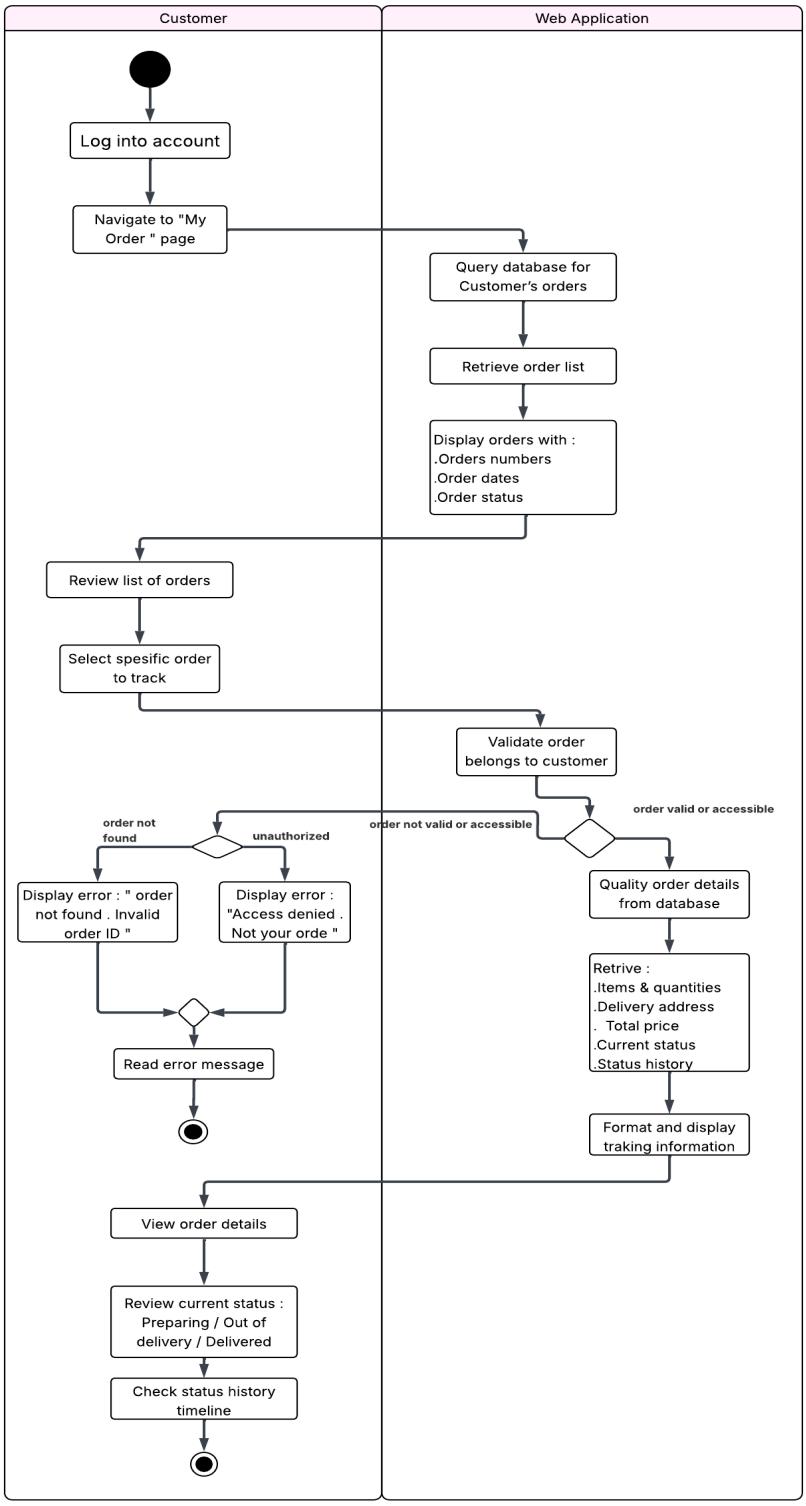
Link : [Add a New Plant to the Catalog]

2. Customer Places an Order [Heba Abu Taha]



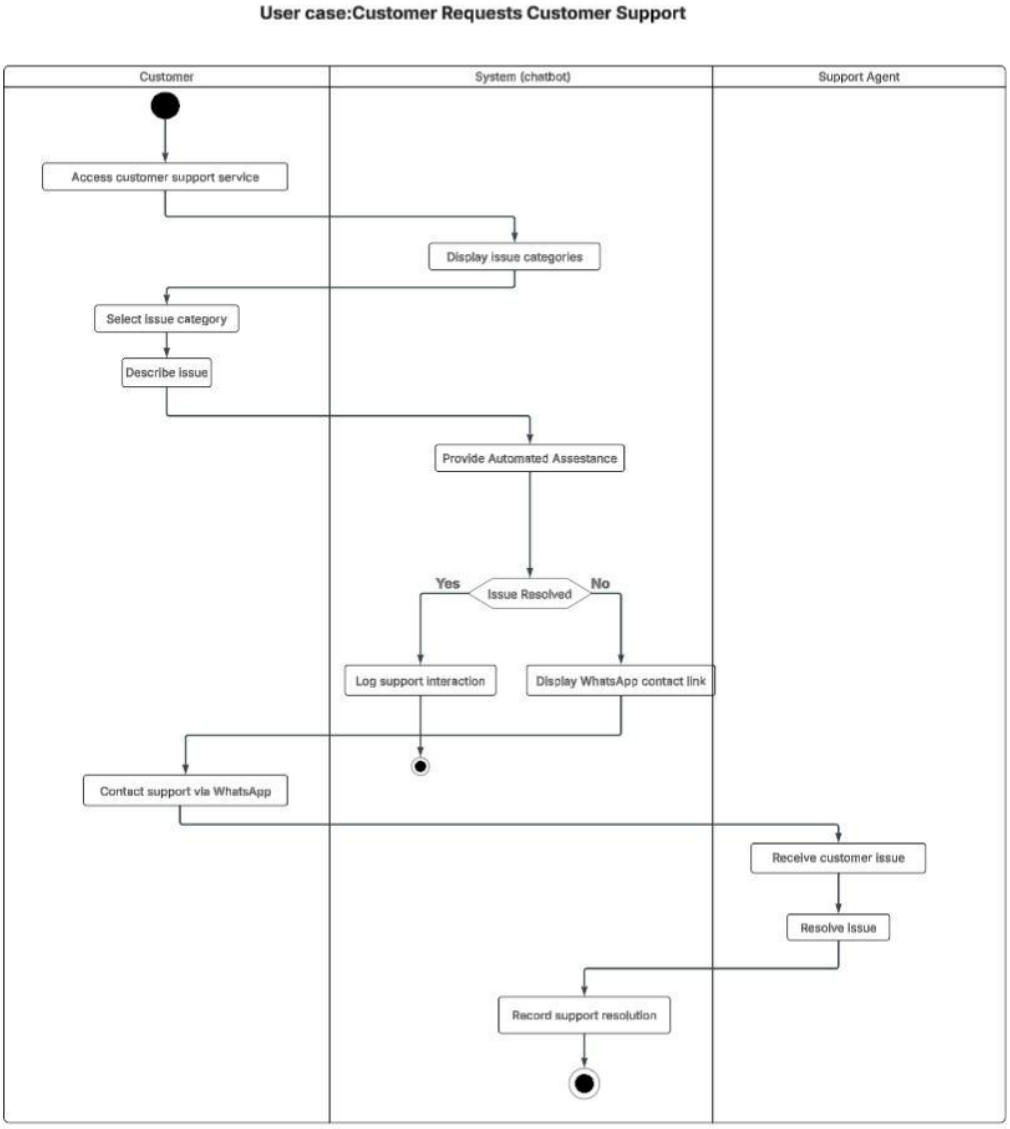
Link : [Customer Places an Order]

Customer Tracks an Order [Individual Task – Duaa Naji] Modified



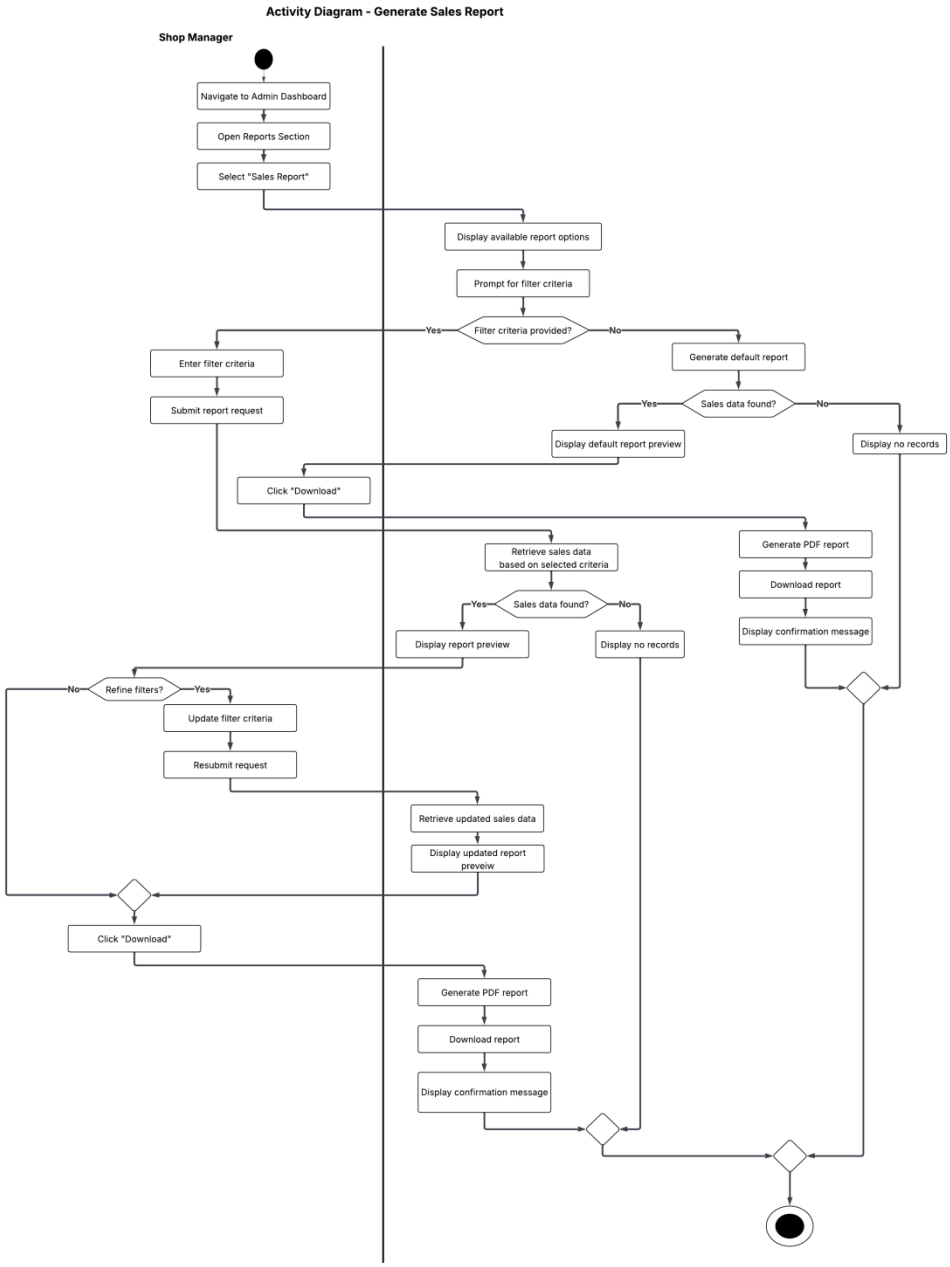
Link : [Customer Tracks an Order]

3. Customer Requests Customer Support [Khadija Absi]



Link : [Customer Requests Customer Support]

4. Shop Manager Generates Sales Report [Individual Task - Jana Quttosa]



Link : [Shop Manager Generates Sales Report]

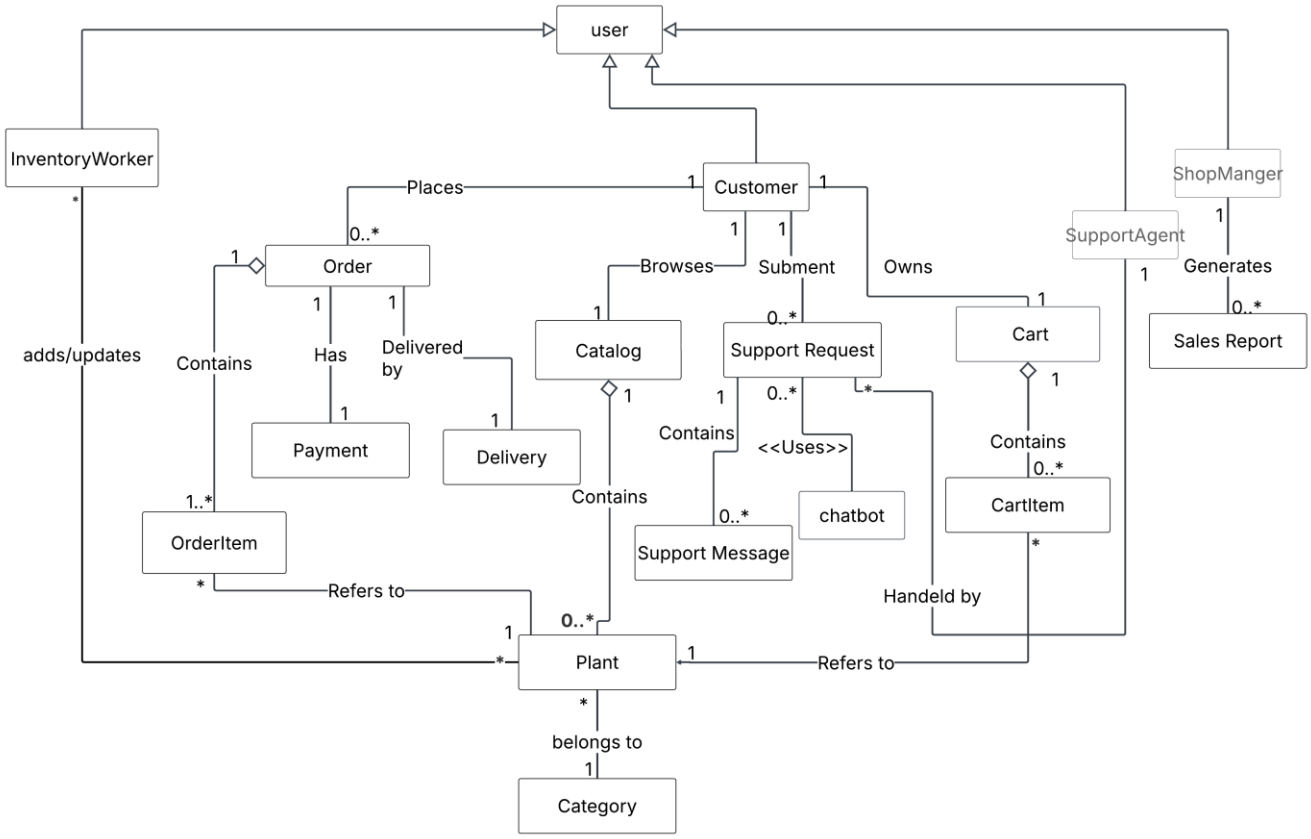
Chapter 3: System Analysis and Modelling

3.1 System CLASS Diagrams

Lead: Enas

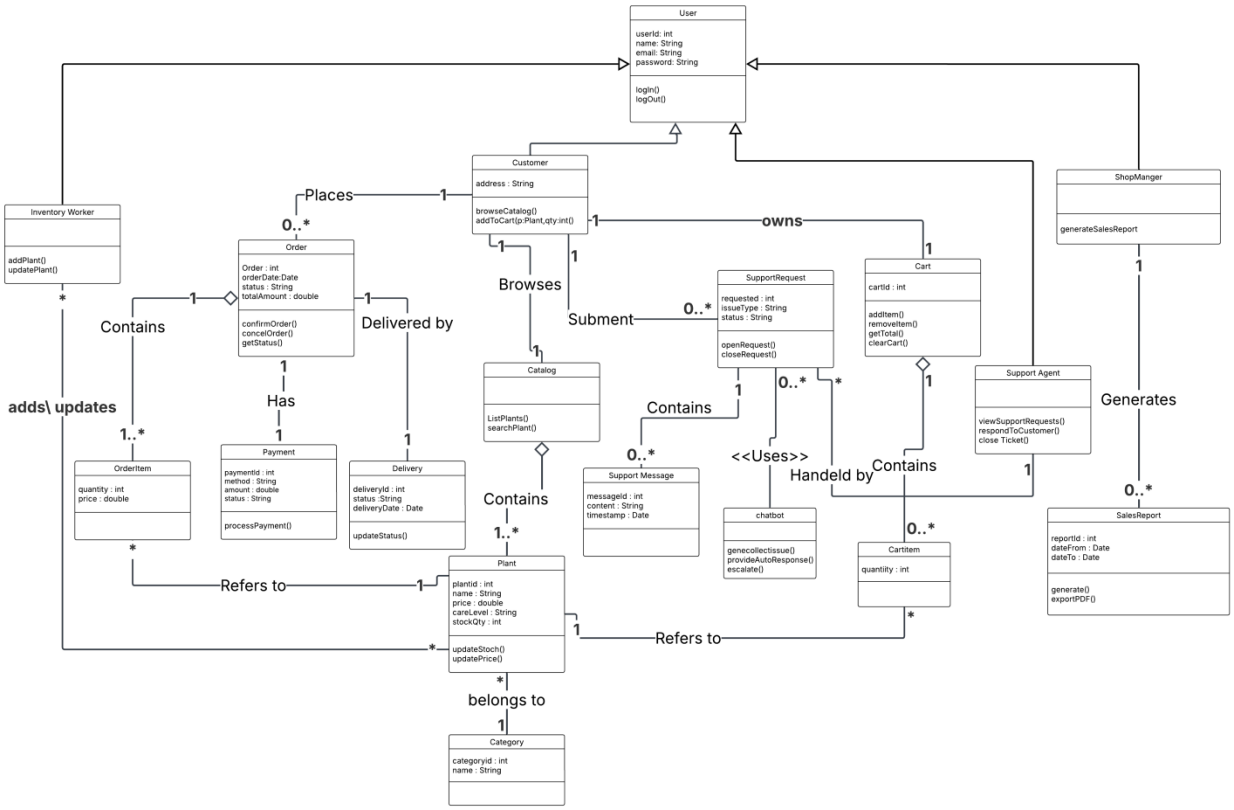
Contributors: Jana: **co-leading,** Heba**: reviewing,** Khadija**: discussing,** Duaa**: finalizing**

Analysis class Diagram



Link : [Analysis class Diagram]

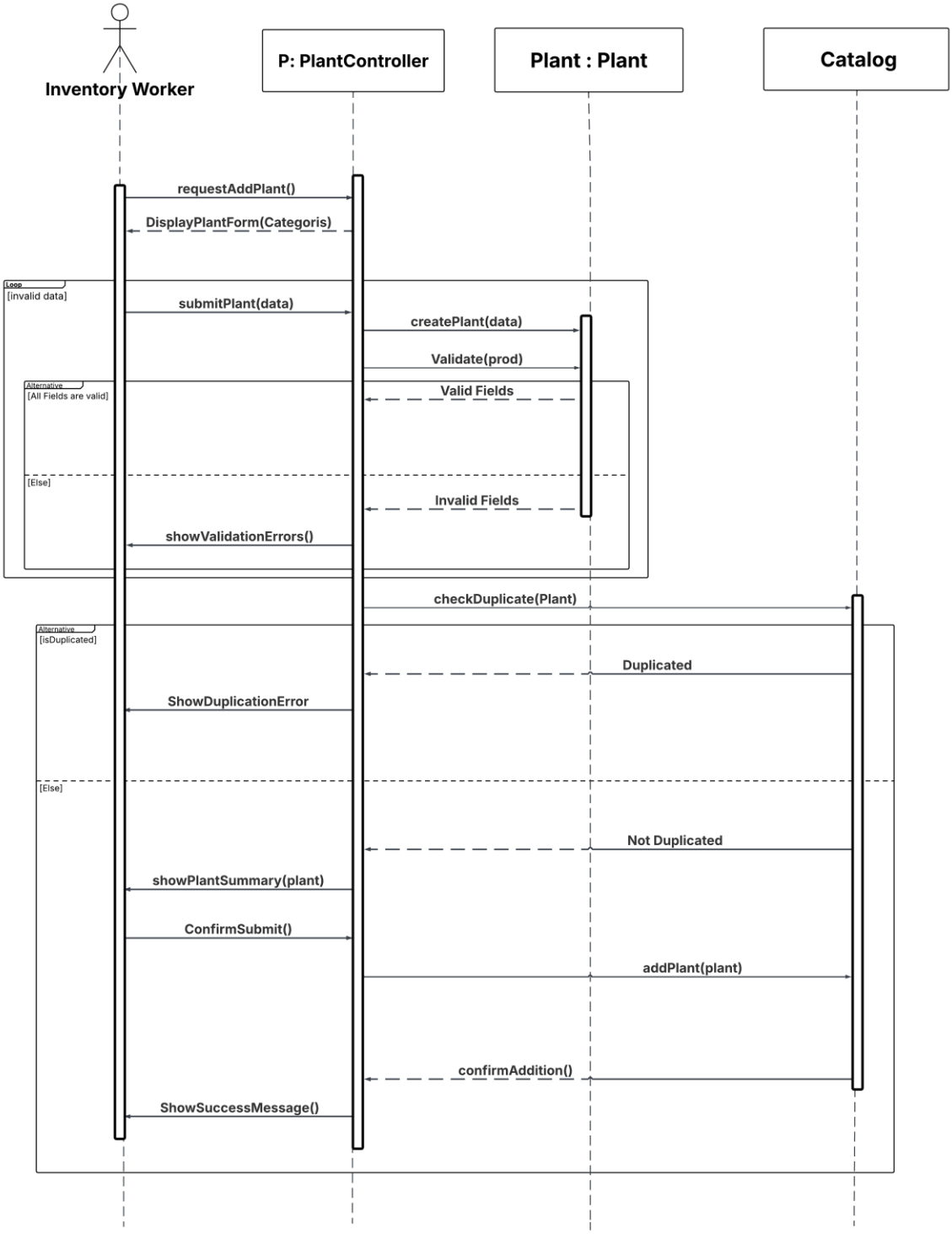
Detailed Class Diagram



Link : [Detailed Class Diagram]

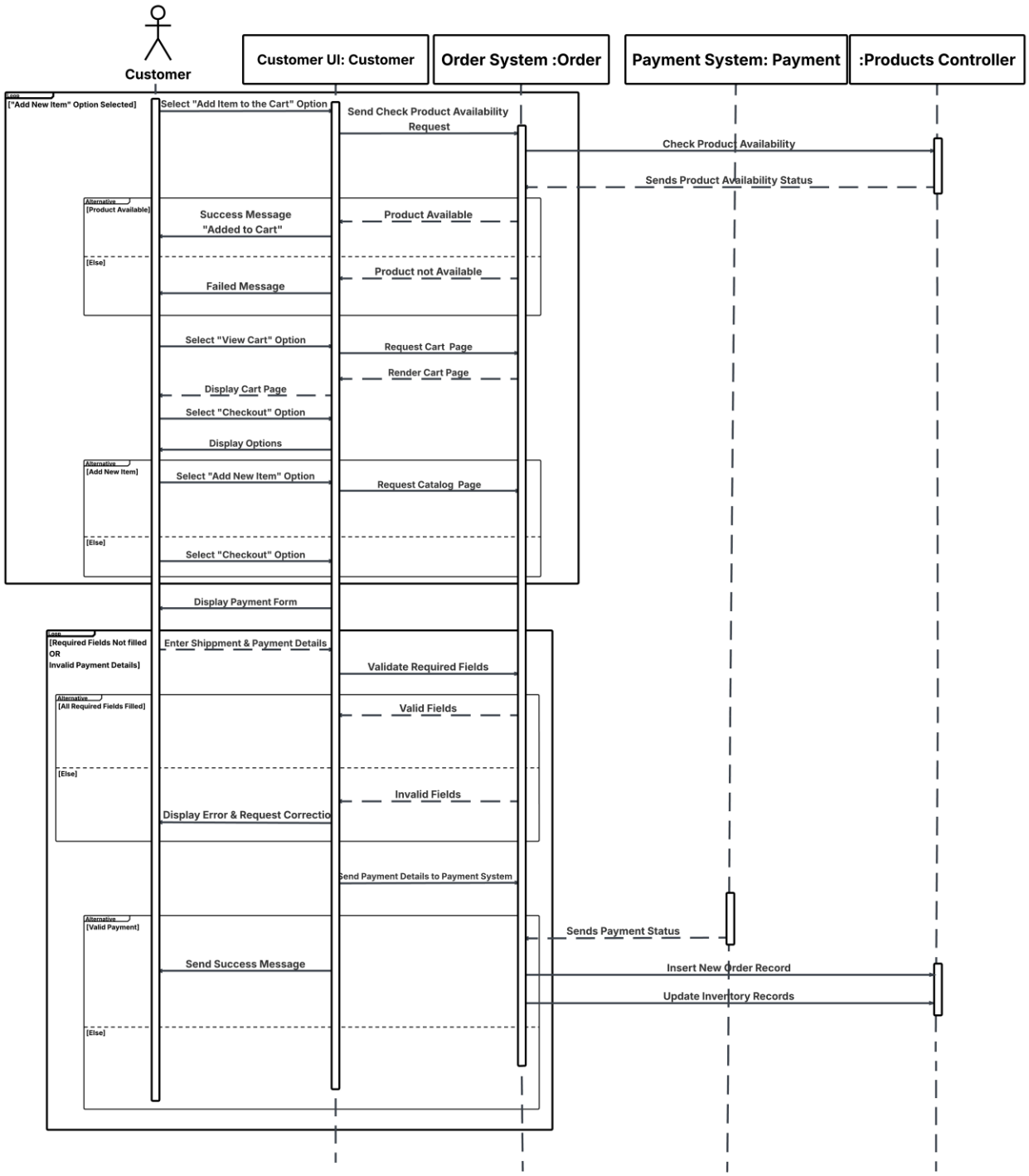
3.2 SEQUENCE Diagram

1. Add a new plant to catalog [Individual Task – Enas Qutit]



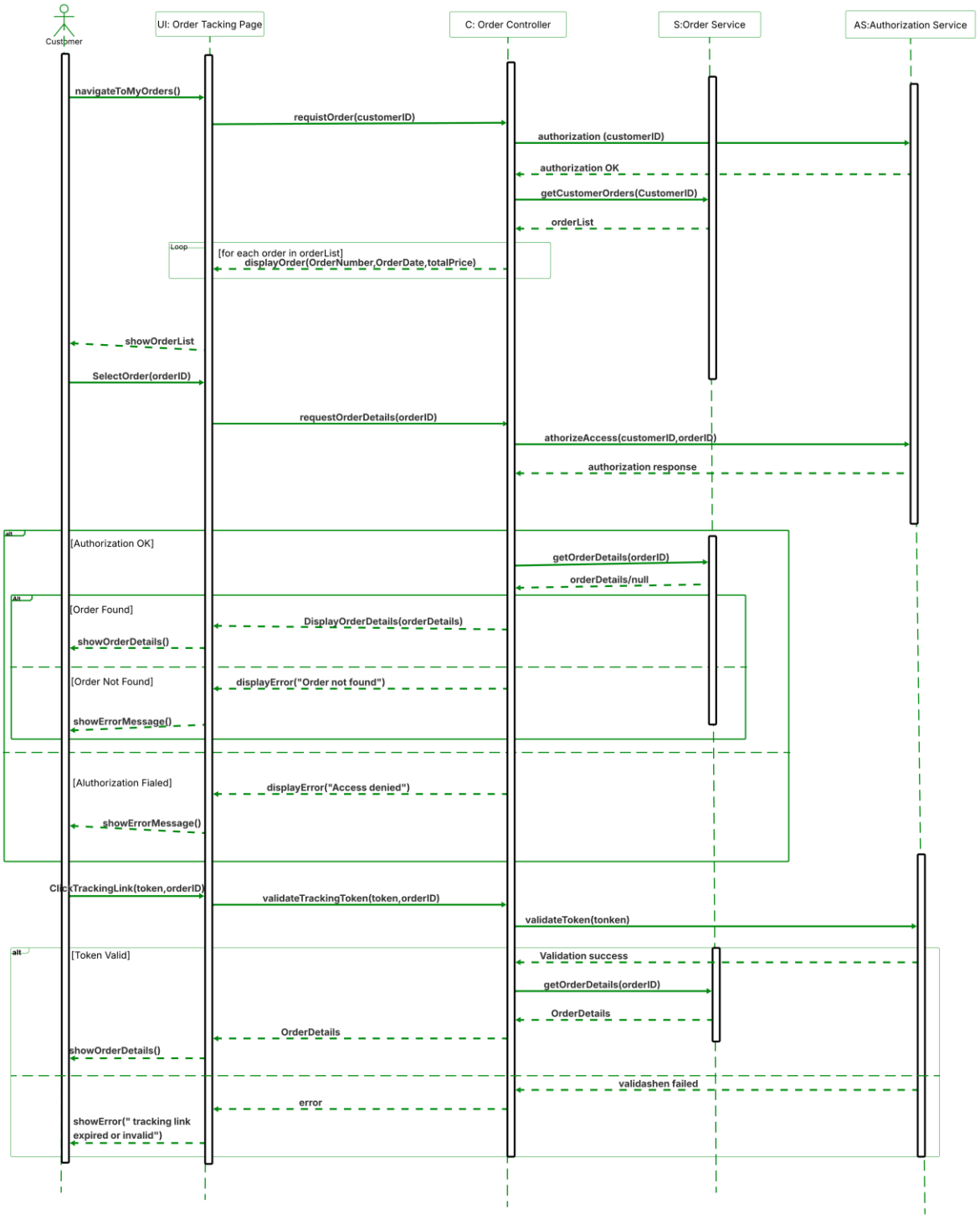
Link : [Add a new plant to catalog sequence diagram]

2. Customer Places an Order [Individual Task – Heba Abu Taha]



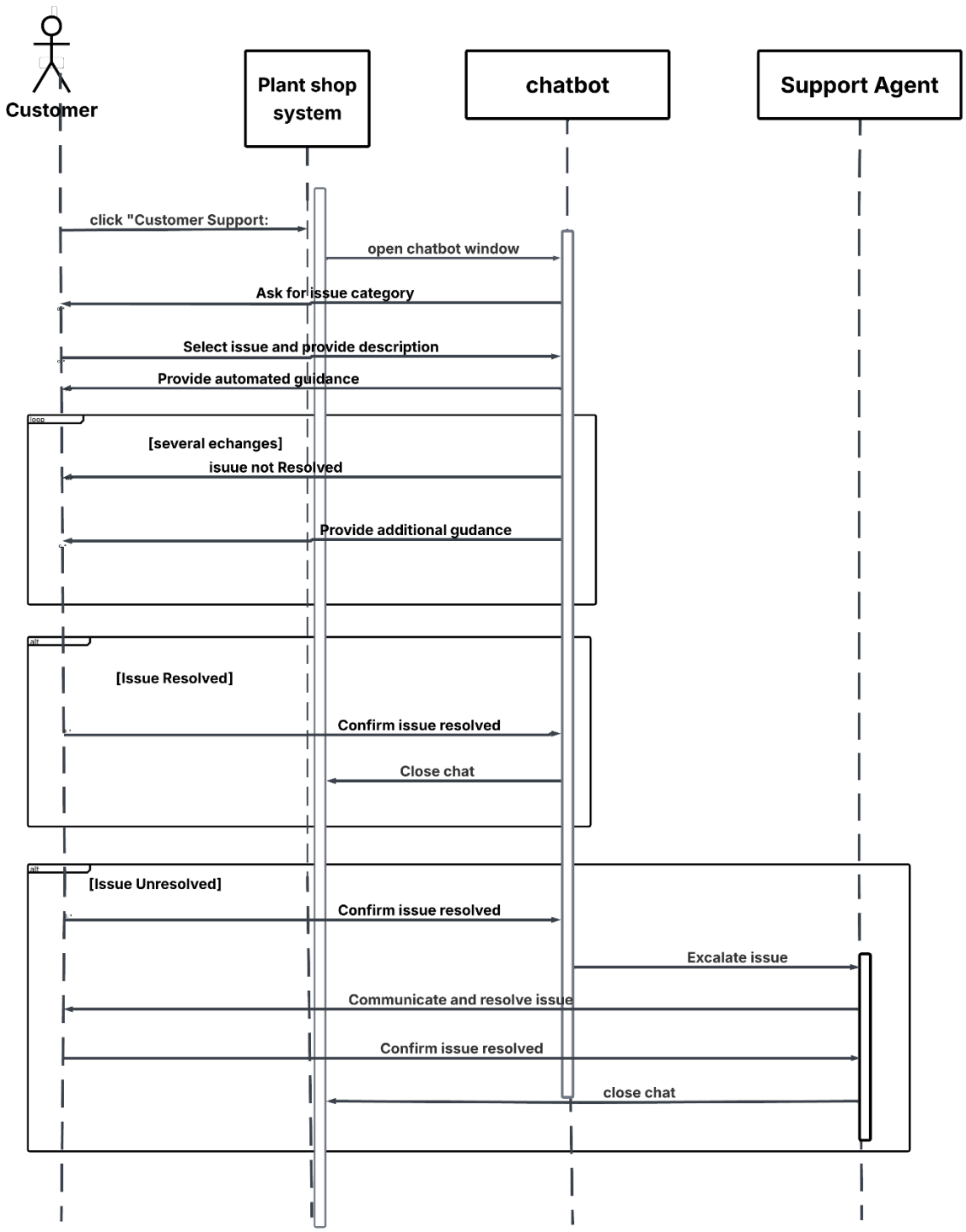
Link : [Customer Places an Order]

3. Customer Tracks an Order [Individual Task – Duaa Naji]



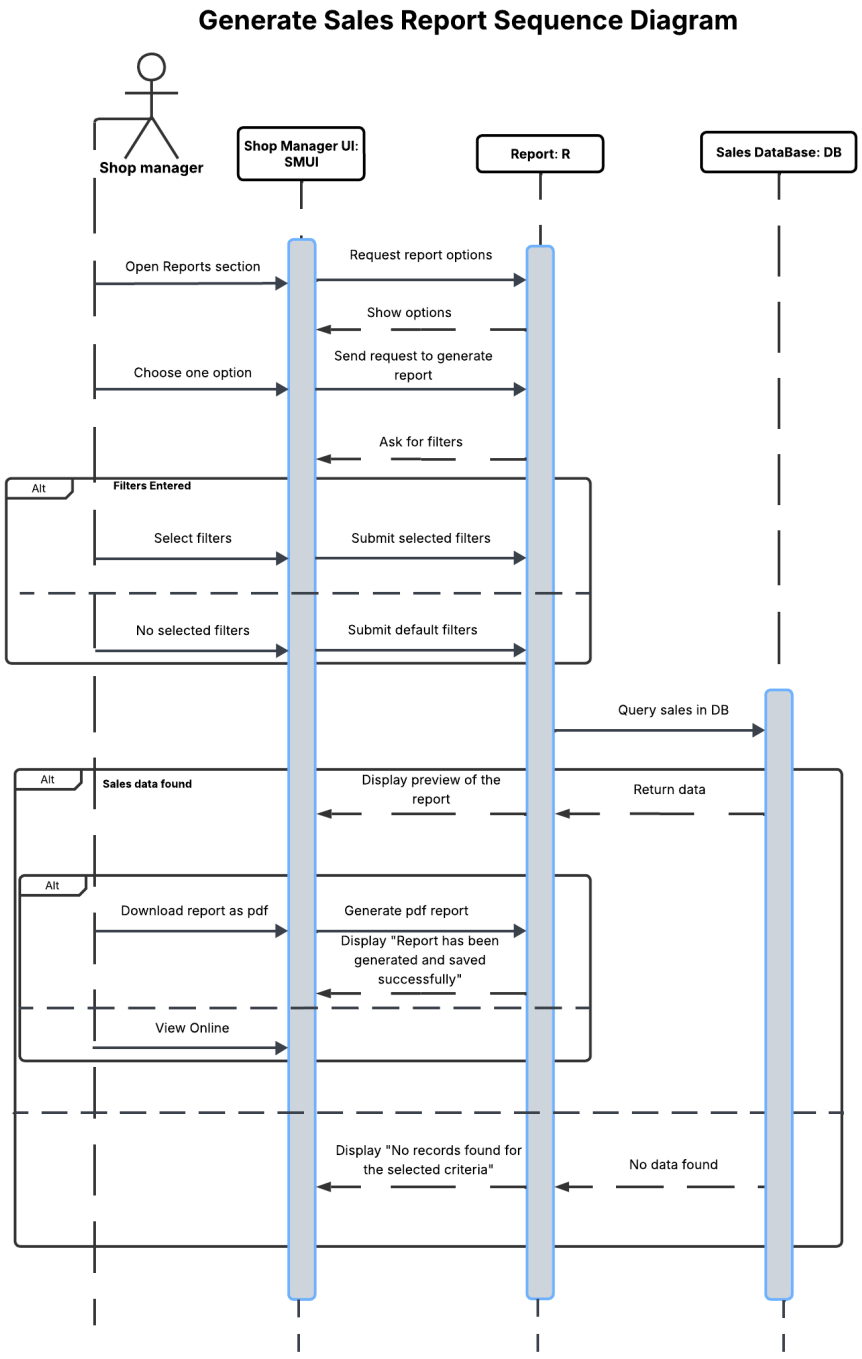
Link : [Customer Tracks an Order]

4. Customer Requests Customer Support [Individual Task – Khadija Absi]



Link : [Customer Requests Customer Support]

5. Shop Manager Generates Sales Report [Individual Task - Jana Quttosa]



Link : [Shop Manager Generates Sales Report]

Chapter 4: System Design and Modelling

4.1 Description of chosen Design Goals

Lead: [Duaa]

Contributors: Khadija: co-leading, Enas: reviewing, Heba: discussing, Jana: finalizing

1. Low Coupling

Low coupling means that each class or component has minimal dependency on others. This allows individual parts of the system to be developed, tested, or modified independently, which improves maintainability and scalability.

How we achieved this:

In the **Class Diagram**, responsibilities are clearly separated. Core classes such as *Customer*, *Plant*, *Order*, *SupportRequest*, and *Payment* are designed as independent entities that do not directly manipulate each other's internal data. Associations exist only where necessary and are defined with clear boundaries, reducing interdependencies between classes.

In the **Component Diagram**, system components such as Order Management, Account Management, Inventory Management, Payment Processing, *and* Support Management are modular and communicate through well-defined interfaces. This design avoids tight integration between components and demonstrates low coupling at the architectural level.

2. High Cohesion

High cohesion means that each class or component focuses on a single, well-defined responsibility and contains only closely related functionalities. This makes the system easier to understand, maintain, and extend.

How we achieved this:

In the **Class Diagram**, each class has a focused purpose. For example, the *Plant* class is responsible only for plant-related attributes such as type, price, availability, and care information, and its associations with Cart, Wishlist, and Order. The Order class manages order-specific data and is linked to *Payment* for transaction handling. This clear focus of responsibilities reflects strong cohesion.

Similarly, in the **Component Diagram**, components such as *Inventory Management* and *Sales Reporting* are dedicated to specific system functionalities. Each component handles a single aspect of the system's behavior, demonstrating high cohesion at the component level.

3. User-Friendly Interface

A user-friendly interface in this system is achieved through clear separation of user interactions and strict role-based access rather than focusing on visual or usability aspects alone. This design approach ensures that each user role interacts only with the functionalities relevant to its responsibilities, which reduces system complexity and minimizes operational errors.

How we achieved this:
In the Component Diagram, separate user interface components are defined for each user role, including Customer UI, Inventory Worker UI, Support Agent UI, and Shop Manager UI. This separation enforces role-based access control and reduces coupling between user interactions and business logic, as each interface communicates with the system through well-defined component boundaries.

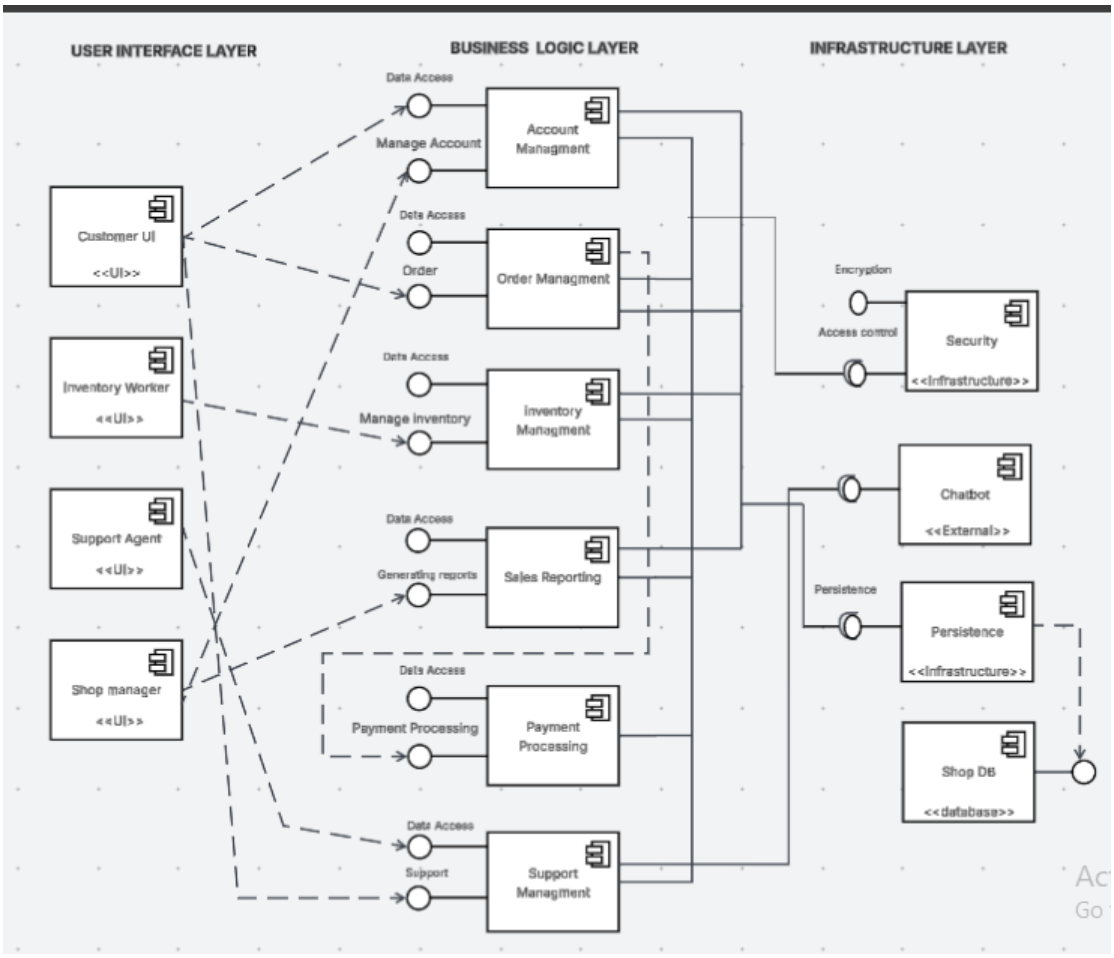
The Class Diagram further supports this design by maintaining clear associations between user-related classes (Customer, SupportAgent, InventoryWorker) and the system entities they interact with, such as Plant, Cart, Order, and SupportRequest. This structure ensures that user actions are consistently mapped to authorized system operations.

Additionally, integrating the Chatbot as a dedicated support component allows customer assistance and issue escalation to be handled independently from core business processes. This separation improves system maintainability and ensures that support functionality does not interfere with primary system operations.

4.2 Component Diagram

Lead: [Enas];

Contributors: Khadija: co-leading, Duaa: reviewing, Heba: discussing, Jana: finalizing



The Component Diagram illustrates a well-structured three-tier architecture that effectively supports key design principles. Low coupling is achieved through clear layer separation and the use of Data Access objects as intermediaries between the Business Logic and Infrastructure layers. Components such as Account Management, Order Management, and Inventory Management operate independently through well-defined interfaces, ensuring that modifications to one component have minimal impact on others.

High cohesion is maintained by grouping related functionalities within focused components. Each component in the Business Logic Layer has a singular purpose—Payment Processing

handles all payment operations, Inventory Management manages stock tasks, and Sales Reporting handles reporting functions exclusively.

User-friendliness and role-based access are addressed through dedicated UI components for different actors. The Customer UI, Inventory Worker UI, Support Agent UI, and Shop Manager UI ensure that each user interacts only with features relevant to their role, reducing complexity and enhancing usability.

The Infrastructure Layer provides centralized services such as Security, Persistence, and Chatbot support, promoting code reusability and consistency across the system. This modular architecture supports maintainability, scalability, and extensibility, enabling the system to evolve efficiently with changing business requirements.

4.3 Overall architecture diagram

Lead: [Khadija];

Contributors: Duaa: co-leading, Enas: reviewing,Jana: discussing, Heba: finalizing

The diagram illustrates the overall architecture of the plant shop system, which is designed following the Layered Architecture approach. The system is organized into three primary layers, each with a clear and well-defined responsibility.

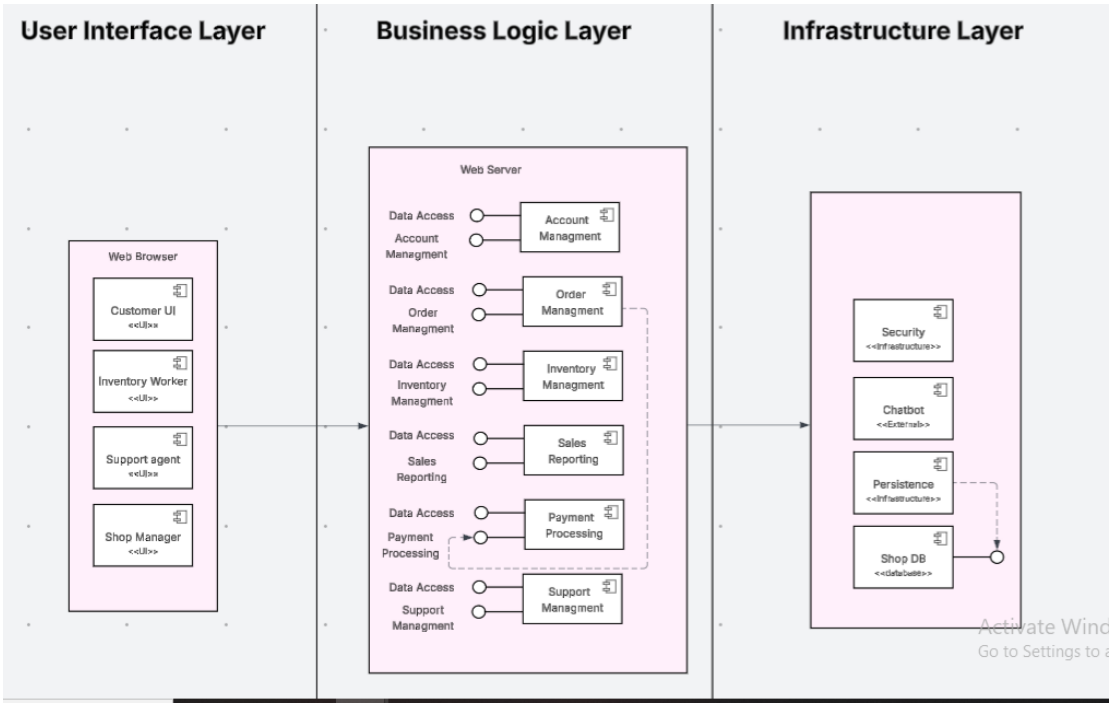
The User Interface (UI) Layer represents the interaction point between users and the system. It provides dedicated user interfaces for different roles, including the Customer, Inventory Worker, Support Agent, and Shop Manager. Each role is provided with a tailored interface that allows users to perform tasks related to their specific responsibilities, ensuring ease of use and role-based access.

The Business Logic (Application) Layer contains the core functionality of the system. It is responsible for processing all user requests and implementing the main business rules. This layer includes components such as Account Management, Order Management, Inventory Management, Sales Reporting, Payment Processing, and Support Management. These components coordinate system operations and act as an intermediary between the user interface and the infrastructure services.

The Infrastructure Layer provides the technical and external services required to support the system's operation. It includes components such as Security, Persistence, Chatbot services, and the Shop Database. This layer is responsible for data storage, secure communication, and integration with external systems such as banking services.

Each component is placed within its appropriate layer based on its functional responsibility, ensuring a clear separation of concerns across the system. The Layered Architecture style was chosen because it supports key design goals and non-functional requirements. By allowing each layer to interact only with adjacent layers, the architecture achieves low coupling, which simplifies system maintenance and future modifications. Additionally, high cohesion is achieved by grouping related functionalities within the same components, such as handling all order-related operations within the Order Management component.

This architectural design also enhances usability, scalability, and maintainability, enabling the system to evolve efficiently while maintaining a clear and organized structure.



4.4 Deployment diagram

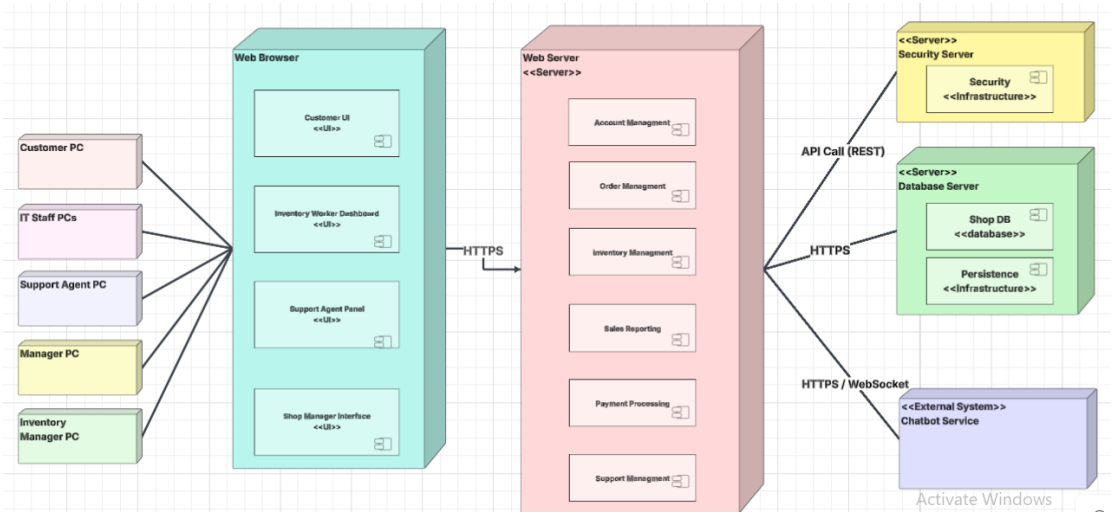
Lead: [Jana];

Contributors: Duaa: co-leading, Khafija: reviewing,Enas: discussing, Heba: finalizing

In the deployment diagram, the system design goals of clear user interaction, low coupling, and high cohesion are clearly addressed. The user interface is accessed through a web browser and supports multiple user roles, allowing each user to interact with the system through a focused and role-specific interface. This separation improves usability and ensures that users access only the functionalities relevant to their responsibilities.

Low coupling is achieved by deploying system functionalities across independent servers that communicate using secure protocols such as HTTPS and RESTful APIs. This design minimizes direct dependencies between system components, allowing each server to operate and evolve independently without affecting other parts of the system.

High cohesion is maintained by grouping related functionalities within the same deployment nodes. For example, all business-related processes are hosted on the web server, while data storage and persistence are handled exclusively by the database server, and security-related services are managed by a dedicated security server. This clear allocation of responsibilities ensures that each component focuses on a specific, well-defined role within the system.



Appendix – I: Minutes of Meetings

Summary of Essential Meetings

Meeting1

Topic: Initiation Phase-Group Formation and Project Planning

Date: Thursday, Nov 6 , 2025

Time: 8:00 PM

Location: Online Meeting (Google Meet)

Attendees:

- Project Manager: Jana Quttosa.
- Team Members: Heba Abu Taha ,Duaa Naji ,Enas Qutit, Khadija Absi.

Meeting topic description

After forming our team, we held a meeting to assign roles, agree on our management strategy, and select our project preferences. Each member was assigned a specific role based on their strengths, and we decided to meet online via Google Meet to discuss future project phases and coordination. We chose Home Plants Online Shop as our preferred business, as it meets the course requirements of being an online system with essential security features.

Meeting 2

Topic: Phase One-Business Definition

Date: Monday, Nov 24 , 2025

Time: 10:00 AM

Location: Online Meeting (Google Meet)

Attendees:

- Project Manager: Jana Quttosa.
- Team Members: Heba Abu Taha ,Duaa Naji ,Enas Qutit, Khadija Absi.

Meeting topic description

As a customer group, we held a meeting to understand the business assigned to our developer group and to prepare the required business description outline. During the meeting, we reviewed the concept of the online cooker shop , discussed its services, business processes, and the roles of employees involved. We also researched similar platforms to better define its operations and capacities. The team collaborated to divide the writing tasks and ensure that the description covered all required aspects, including services offered, user roles, and how the business functions. This meeting helped us align our understanding and organize the preparation of the document to be submitted and shared with the developer group.

Meeting 3

Topic: Phase Two-Requirements Engineering

Date: Friday, Dec 12 , 2025

Time: 5:00 PM

Location: Online Meeting (Google Meet)

Attendees:

- Project Manager: Jana Quttosa.
- Team Members: Heba Abu Taha ,Duaa Naji ,Enas Qutit, Khadija Absi.

Meeting topic description

Our team held a meeting to plan and organize the work required for Phase Two: Requirement Engineering. We began by reviewing the customer's business description to ensure a shared understanding of the system goals. We then discussed how to approach each of the required tasks in this phase: User Requirements (Task 2.1), System Requirements (Task 2.2), and Effort and Time Estimation (Task 2.3). During the meeting, we assigned roles, selected task leaders, and agreed on a strategy for collaboration, including how each member would contribute (e.g., writing, reviewing, finalizing). We also decided on timelines for drafting and reviewing each task to ensure we stay on track for submission. This meeting helped us align

responsibilities, clarify the scope of each task, and prepare for completing the phase efficiently and on time.

Meeting 4

Topic: Phase Three-Requirement Analysis and Modelling

Date: Saturday, Dec 27 , 2025

Time: 8:00 PM

Location: Online Meeting (Google Meet)

Attendees:

- Project Manager: Jana Quttosa.
- Team Members: Heba Abu Taha ,Duaa Naji ,Enas Qutit, Khadija Absi.

Meeting topic description

In this meeting, our team discussed the planning and task distribution for Phase 3, which focuses on Scenario Analysis, Actor Identification, Use Case Modeling, and Activity Modeling. We began by reviewing the instructions for each task and identifying which parts are individual and which are group-based. For the individual tasks (Scenario Analysis, Use Case Specification, and Use Case Activity Modeling), each member selected one service of the system to focus on and committed to writing one complete scenario and developing both its detailed use case description and corresponding activity diagram. For the group tasks (Actor Analysis, Use Case Diagram, and Main Activity Diagram), we assigned a leader for each task and agreed on roles for contribution such as reviewing, redrawing, and finalizing. We also agreed to use a single UML tool across the team for consistency in diagramming. This meeting helped us divide responsibilities clearly, ensure all services and actors are covered, and set a timeline(Monday,29Dec, at 6:00PM) for completing this phase efficiently.

Meeting 6,7,8:

Topic: Final Group Project .

Date: Saturday , Jan 24, 2026 & Sunday, Jan 25, 2026&Monday ,Jan 26 ,2026 .

Time: 9:00PM 8:30PM & 9:00PM.

Location: Online Meeting (Google Meet)

Attendees:

- Project Manager: Jana Quttosa.
- Team Members: Heba Abu Taha ,Duaa Naji ,Enas Qutit, Khadija Absi.

Meetings topic description :

Our team held a meeting to plan and prepare for the final project presentation as required in Phase 4. The meeting focused on organizing our slides, dividing presentation parts equally among members, and ensuring each section of the project is clearly represented. We reviewed the format provided, which includes slides for our group roles, business description, selected scenario, use case diagrams, activity and class models, sequence diagram, design goals, component, architectural, and deployment diagrams. Each member selected the part they would present, ensuring equal speaking time and alignment with their individual contributions. We also discussed the time limit of 12 minutes and scheduled rehearsals to practice and stay within the allocated time. The goal of this meeting was to finalize the content, structure, and responsibilities for a smooth and well-coordinated group presentation.

Also , in this final meetings, our team came together to organize and finalize the full project report and presentation as required for submission. We reviewed all phases of the project, gathered the final versions of all diagrams and written tasks, and ensured that feedback received during previous phases and the presentation was properly addressed and integrated. Each member confirmed their individual contributions, and group task leaders verified that contributor names were clearly noted on each task or diagram, as instructed. We also coordinated the creation of the final PDF file, including team member photos and all required chapters, structured as per the Software Requirement Specification and Design (SRSD) format. Additionally, we finalized and reviewed the presentation slides, ensuring that each member was assigned a part to present. The meetings helped ensure that both the final report and presentation were complete, consistent, and ready for timely submission

Meeting with customer

Developer Group: Group8

Customer Group: Group 7

Date: Sunday, Dec 7, 2025

Time: 9:00 PM

Location: Online Meeting (Google Meet)

Attendees:

Developer Group: Duaa Naji, Khadija Absi.

Customer Group: Mariam Dawoud, Waseem Sesan, Abdelrahman Imhimid.

Discussed Topics: During our meeting with the customer group for the Home Plants Shop project, several important improvements were added to the requirements.

The customers requested a communication channel in the form of a WhatsApp link to provide after-sales support. They also wants to improve the order tracking system by assigning distinct colors to each order status for better visual recognition. Additionally, new delivery pricing rules were introduced: orders exceeding 150 ILS will receive a 50% discount on delivery fees, while express deliveries will incur an extra charge and will not be eligible for discounts. It was also specified that express orders placed late in the day will not be delivered on the same day.

Finally, the customers asked for a live chat feature on the website, including an FAQ section, and requested that each plant's gallery include multiple photos showing its appearance across different seasons of the year.